

- By the defn. of conditional, $(p \rightarrow q) \wedge (q \rightarrow r)$ is True and $(p \rightarrow r)$ is False.
- By the defn. of conditional, p is True, r is False.
- By the defn. of AND, $(p \rightarrow q)$ is True and $(q \rightarrow r)$ is True.
- Since r is False and $(q \rightarrow r)$ is True, by the defn. of conditional, q is False.
- Since p is True and q is False, by the defn. of conditional, $(p \rightarrow q)$ is False.

This is a contradiction. $\therefore$ It is a tautology

c) $[p \wedge (p \rightarrow q)] \rightarrow q$

Suppose, for sake of contradiction, it is not a tautology.

$\therefore$ There exists an assignment of truth values to p, q such that

$[p \wedge (p \rightarrow q)] \rightarrow q$ is False. By the defn. of conditional, $p \wedge (p \rightarrow q)$ is True and the q is False. By the defn. of AND, p is True and $(p \rightarrow q)$ is True.

Since q is False and $(p \rightarrow q)$ is True, by the defn. of conditional, p is False.

This is a contradiction. $\therefore$ It is a tautology

d) $[(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow r$

Suppose, for sake of contradiction, it is not a tautology.

$\therefore$ There exists an assignment of truth values to p, q, r such that

$[(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow r$ is False.

By the defn. of conditional, $[(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)]$ is True and r is False.

By the defn. of AND, $(p \vee q)$ is True, $(p \rightarrow r)$ is True and $(q \rightarrow r)$ is True.

By the defn. of conditional, since r is False and $(p \rightarrow r)$ is True and $(q \rightarrow r)$ is True, both p and q are False. By the defn. of OR, $(p \vee q)$ is False.

This is a contradiction. $\therefore$ It is a tautology

15) Show that each conditional statement in Exercise 11 is a tautology ~~using~~ by applying a chain of logical identities as in Example 8. (Do not use truth tables.)

a) $(p \wedge q) \rightarrow p \equiv \neg(p \wedge q) \vee p$ (By conditional-disjunction equivalence)
 $\equiv (\neg p \vee \neg q) \vee p$ (By 1st De Morgan's law)